NI TB GENOMIC SURVEILLANCE

Proof of Concept System

Step-by-Step User Guide

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This guide provides current instructions for deploying, configuring, and operating the TB Genomic Surveillance PoC system on Windows using the updated workflow. It now also explains the synthesis layer that combines analysis outputs for review.

TABLE OF CONTENTS

1. SYSTEM REQUIREMENTS

Operating System: Windows 10 or later

Python: Python 3.10 or later

PostgreSQL: PostgreSQL 12 or later

RAM: Minimum 4 GB (8 GB recommended)

Disk Space: 500 MB for application + 1 GB for data

Network: Localhost access only (development mode)

2. INSTALLATION & SETUP

Step 2.1: Clone or Extract Project

Extract the TB-Genomics project folder to your desktop.

```
C:\Users\YourUsername\Desktop\TB-Genomics-main
```

Step 2.2: Create Python Virtual Environment

Open PowerShell in the project directory and create a Python virtual environment:

```
python -m venv .venv
```

Step 2.3: Activate Virtual Environment

Activate the virtual environment in PowerShell:

```
.\.venv\Scripts\Activate.ps1
```

You should see (.venv) at the start of the command line.

Step 2.4: Install Python Dependencies

With the virtual environment active, install required packages:

```
pip install -r backend/requirements.txt
```

This installs FastAPI, SQLAlchemy, matplotlib, networkx, scipy, reportlab, and other dependencies.

Step 2.5: Preferred Startup Method (One-Click .bat Files)

For most users, the easiest and recommended startup method is the provided Windows batch files:

```
Setup-And-Start-Platform.bat
```

First-time setup: creates .venv (if needed), installs dependencies, starts backend + GUI.

```
Start-Platform.bat
```

Normal use: starts backend + GUI.

```
Start-Backend.bat
```

Backend-only mode for API testing or separate GUI hosting.

Sections 4 and 5 below describe manual startup commands if batch launch is not available.

3. DATABASE CONFIGURATION

Step 3.1: Start PostgreSQL Service

Ensure PostgreSQL is running. Open Services (services.msc) and verify that 'postgresql-x64' service is started.

Step 3.2: Connect to PostgreSQL

Open PowerShell and connect to PostgreSQL. If psql is not on PATH, use the full executable path:

```
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U postgres -h localhost -d
postgres
```

Enter your PostgreSQL password when prompted.

Step 3.3: Create Database User & Database

In psql, run these commands:

```
CREATE USER tb WITH PASSWORD 'tb';
CREATE DATABASE tb_surveillance OWNER tb;
GRANT ALL PRIVILEGES ON DATABASE tb_surveillance TO tb;
```

Step 3.4: Load Database Schema

Exit psql (type \q) and load the schema. In PowerShell use the -f flag instead of shell redirection:

```
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U tb -h localhost -d
tb_surveillance -f .\db\schema.sql
```

When prompted for password, enter 'tb'.

If psql is already on PATH, you can remove the full executable path and keep the same arguments.

4. BACKEND SERVER CONFIGURATION

If you started the platform via Start-Platform.bat or Start-Backend.bat, you can skip this section.

Step 4.1: Start Backend Server

In PowerShell (with virtual environment active), start the FastAPI backend server:

```
uvicorn backend.app:app --host 0.0.0.0 --port 8000 --reload
```

You should see output indicating the server is running on http://0.0.0.0:8000

Step 4.2: Verify Backend Health

Open your browser and visit:

```
http://localhost:8000/docs
```

You should see the FastAPI interactive API documentation page.

Note: Leave this terminal running. Open a new PowerShell window for the next steps.

5. FRONTEND SERVER SETUP

If you started the platform via Start-Platform.bat, the GUI is already running and this section can be skipped.

Step 5.1: Navigate to GUI Directory

In a new PowerShell window, navigate to the GUI directory:

```
cd C:\Users\YourUsername\Desktop\TB-Genomics-main\gui
```

Step 5.2: Start Frontend Server

Start the Python HTTP server on port 8081:

```
python -m http.server 8081
```

Step 5.3: Access the Application

Open your browser and visit:

```
http://localhost:8081
```

You should see the NI TB Genomic Surveillance interface with workflow steps for ingest, analysis, search, and governance.

6. GENERATING SYNTHETIC DATA

Safety note: synthetic seeding is disabled by default to protect operational use.

To enable demonstration mode in the backend PowerShell session, set:

```
$env:TB_ENABLE_SYNTHETIC_SEEDING="1"
```

If you also need to run full analysis/reporting on synthetic data, set:

```
$env:TB_ALLOW_NON_OPERATIONAL_ACTIONS="1"
```

For real operational use, leave these variables unset.

Step 6.1: Using the Web Interface

In the browser on http://localhost:8081:

1. Navigate to **Step 2 — Upload data**
2. Set **Cases: 250** (or desired number)
3. Set **Seed: 42** (for reproducible results)
4. Check **Reset existing synthetic data first**
5. Click **Generate synthetic dataset**
6. Wait for confirmation message

What Happens:

- The system fetches current TB incidence data from the World Bank API
- 250 realistic synthetic TB cases are generated using country-weighted distribution
- Each case is assigned lineage, resistance profile, and location
- Cases are organized into 4-6 outbreak clusters
- All data is pseudonymised and stored in the database

7. RUNNING ANALYSIS JOBS

Step 7.1: Run Clustering Analysis

In **Step 3 — Run analysis**:

1. Click **Run clustering**
2. A progress bar will appear and update as the job runs
3. When complete, status shows **Completed**

Result: Cases are analyzed for genetic similarity and grouped into clusters.

Step 7.2: Generate Outbreaker2 Inputs

In **Step 3 — Run analysis**:

1. Click **Generate outbreaker2 inputs**
2. Wait for completion

Result: Case data and DNA sequences are prepared for outbreak investigation analysis.

Step 7.3: Run Outbreaker2 Analysis

In **Step 3 — Run analysis**:

1. Click **Run outbreaker2**
2. This may take 30-60 seconds
3. Status will show when analysis completes

Result: Outbreak analysis completes and generates diagnostic plots, an enhanced transmission network, and synthesis-ready review outputs for the investigation team.

Note: If R/outbreaker2 is unavailable, the system falls back to a mock report generator so the workflow remains usable for demos.

8. SYNTHESIS LAYER AND INTERPRETING RESULTS

The synthesis layer sits between analytics and investigation. It combines sequence-cluster evidence, Outbreaker2 posterior links, specimen dates, region, lineage, resistance patterns, and cluster context into a review summary.

Important: the synthesis scores, flags, and priority values are heuristic and non-validated. They are designed to support human review, not replace sign-off or expert judgement.

What the synthesis layer gives you:

- **Transmission confidence:** High, moderate, low, or insufficient evidence
- **Contradiction flags:** Warnings for mismatched genomic, timing, geography, or resistance signals
- **Priority score:** A ranked review score that helps staff focus on the most important clusters or pairs first
- **Recommended review actions:** A short list of next steps for the investigation team

Common flags you may see:

- High posterior link but large genomic distance
- Low genomic support with no clear epidemiological or geographic link
- Missing sequence or QC data
- Wide date spread inside a single cluster
- Cluster spans multiple regions
- Resistance signal appears inside the cluster

Step 8.2: View Case Summary

In **Step 4 — Results**:

1. Click **View cases**

2. A raw JSON view of all cases is displayed

Include fields: case ID, specimen date, region, lineage, resistance profile

Step 8.3: View Outbreak Analysis Plots

In **Step 4 — Results**:

1. Click **View outbreaker2 summary**

2. The results panel displays:

- **Case Summary:** Total cases, clustered cases, unclustered cases
- **Outbreak Analysis:** MCMC analysis status and likelihood statistics
- **Transmission Network Insights:** Node count, high-confidence links, and ranked priority spreaders
- **Diagnostic Plots:**
  - **MCMC Trace:** Shows convergence of likelihood estimate
  - **Posterior Distributions:** 4-panel histogram of key parameters
  - **Transmission Tree:** Network diagram showing inferred transmission chains
  - **Phylogenetic Tree:** Genetic similarity clustering of isolates
  - **Resistance Heatmap:** Drug resistance pattern overview across cases

Step 8.4: Download the Outbreak Report PDF

In **Step 4 — Results**:

1. Click **Download outbreak report (PDF)**
2. Open the generated report
3. The updated report preserves image aspect ratios so plots are not stretched or squashed

Step 8.5: View Governance & Audit Trail

In **Step 5 — Governance & Compliance**:

1. Click **View audit trail**
2. A detailed log of all system actions is displayed with timestamps

This provides accountability for all data processing and analysis operations.

9. TROUBLESHOOTING

■ ***Browser cannot connect to backend***

✓ Ensure backend is running (uvicorn command). Check firewall settings. Verify port 8000 is available.

■ ***Database connection error***

✓ Verify PostgreSQL is running. Check user 'tb' exists with password 'tb'. Ensure database 'tb_surveillance' was created.

■ ***psql command is not recognized***

✓ Use the full PostgreSQL executable path or add the PostgreSQL bin directory to PATH. In PowerShell, load schema with `psql -f .\db\schema.sql`, not `< db/schema.sql`.

■ ***Synthetic data generation fails***

✓ Check internet connection (World Bank API is called). If offline, system uses fallback values. Verify database is accessible.

■ ***Analysis jobs timeout or fail***

✓ Check disk space in exports/ folder. Verify all Python dependencies are installed. Try regenerating synthetic data.

■ ***Graphics not displaying***

✓ Ensure matplotlib is installed. Check exports/ folder contains PNG files. Try refreshing browser (Ctrl+F5).

■ ***PDF images look distorted***

✓ Regenerate the outbreak report from the updated backend. The current report code scales images proportionally to preserve aspect ratio.

NEXT STEPS & SUPPORT

After PoC Validation:

- Deploy to test environment
- Connect to real TB genomic data sources
- Review synthesis-layer thresholds with public health and laboratory leads
- Configure authentication (LDAP/AD integration)
- Set up database backup and recovery procedures
- Deploy Outbreaker2 R package on analysis servers
- Implement production monitoring and logging

Support Resources:

- FastAPI Documentation: https://fastapi.tiangolo.com
- Outbreaker2 Package: https://www.repidemicsconsortium.org/outbreaker2
- PostgreSQL Documentation: https://www.postgresql.org/docs